

# GP-Enhanced Sparse Regression for Financial PDE Discovery from Option Market Data

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## Abstract

We discover governing partial differential equations for option pricing directly from market data using sparse regression on derivative-form libraries (SINDy). The core bottleneck in financial PDE discovery is that the relevant data — option price surfaces — are noisy, and the canonical Black–Scholes library requires second-order derivatives in the strike direction, which finite differences amplify catastrophically. We replace finite-difference and Savitzky–Golay derivatives with *Gaussian Process* (GP) derivatives obtained analytically from a learned RBF/Matern kernel, and systematically compare six derivative-estimation strategies (FD, SavGol, neural/autograd, spectral/FFT, weak-form, GP) under noise. GP achieves  $R^2(\text{clean}) = 0.990$  at 1% noise and remains the best classical estimator through 10% noise ( $R^2 = 0.920$  vs. 0.816 for SavGol). A hard-constraint PINN reduces the put-option relative  $L^2$  error from 48.0% to 4.98%, validating the discovered Black–Scholes structure. On 1,374 real option contracts (SPY/QQQ/AAPL/MSFT, May 2026 chains) GP-SINDy recovers  $R^2 = 0.966$  on SPY and identifies the dominant first-order terms; an SPY contribution analysis attributes 94.5% of the dynamics to two terms, consistent with the BS first-order structure once standardized. Misspecification probes on Merton jump-diffusion data flip a known-zero diffusion coefficient to 1.90, demonstrating SINDy’s value as a model-diagnostic tool.

## 1 Introduction

Discovering governing equations from data is a now-mature program in dynamical systems [2, 20, 21]. Financial applications are conceptually attractive — we observe rich option surfaces in  $(S, K, T)$  that should obey known PDEs (Black–Scholes [1], Dupire [6], Heston [12]) — but practically hard: option prices are noisy, the canonical library uses second derivatives in strike, and the BS coefficient vector is multicollinear ( $S\partial_S V$  and  $V$  are nearly aligned on log-moneyness grids). The closest prior work, Feng et al. [8] (NeurIPS 2025 Workshop), recovers *stochastic* BSDEs from single-stock AAPL trajectories via stochastic SINDy under the risk-neutral measure.

This work takes a complementary route: deterministic PDE discovery from *cross-sectional* option surfaces. The key methodological move is to replace finite-difference and Savitzky–Golay derivatives, which dominate the SINDy literature, with derivatives obtained analytically from a Gaussian Process regression on the price surface. This change moves the practical noise tolerance of financial PDE discovery from  $< 1\%$  to  $\sim 10\%$ , while remaining as cheap as classical SINDy ( $\sim 10$  s per noise level vs.  $\sim 135$  s for a hard-constraint PINN). Our contributions are:

1. **GP-derivative SINDy for financial PDEs.** We use analytical derivatives of a learned RBF / Matern GP as library columns. To our knowledge this is the first application of GP-derivative SINDy to option-price surfaces.
2. **Systematic 6-method comparison.** We evaluate FD, SavGol, neural/autograd, spectral/FFT, weak-form, and GP on a common 13-point noise grid using  $R^2(\text{clean})$  — the  $R^2$  of the time-derivative residual against *clean* ground truth, separating fit quality from coefficient accuracy.

3. **PINN validation with hard constraints.** A hard-constraint PINN that bakes the put payoff into the trial function reduces relative  $L^2$  from 48.0% to 4.98% ( $R^2 : 0.599 \rightarrow 0.996$ ), validating the discovered PDE.
4. **Real-market discovery.** On 1,374 contracts (SPY/QQQ/AAPL/MSFT, May 2026) we obtain GP-SINDy  $R^2$  up to 0.966 on SPY, with a contribution analysis isolating the dominant terms, and a working IV-smile / term-structure recovery.
5. **Misspecification diagnostic.** On synthetic Merton jump-diffusion data, the SINDy  $d^2V/dS^2$  coefficient (true value 0) is identified as 1.905 — a positive-control test showing SINDy can flag missing physics.

## 2 Method

**Problem setup.** Given a noisy observation of the option surface  $V(S, t)$  on a grid, we wish to discover a PDE of the form  $\partial_t V = \sum_k \theta_k \phi_k(V, S)$  where  $\{\phi_k\}$  is a candidate library. Our library  $\Phi$  contains  $\{V, \partial_S V, \partial_{SS} V, S \partial_S V, S^2 \partial_{SS} V\}$ ; the true BS coefficients are  $(rV, \cdot, \cdot, -rS \partial_S V, -\frac{1}{2}\sigma^2 S^2 \partial_{SS} V)$  with  $r = 0.05, \sigma = 0.2$ , giving the target  $(0.05, 0, 0, -0.05, -0.02)$ . Standardized least-squares with sequentially-thresholded least squares (STLS) is used throughout.

**GP derivative estimation.** We fit  $V(S, t) \sim \mathcal{GP}(0, k(\cdot, \cdot))$  with an anisotropic RBF (synthetic) or Matern-3/2 (real-data) kernel. Derivatives are obtained analytically:  $\partial_S V$  and  $\partial_{SS} V$  are conditional means of derivative-augmented kernels, avoiding numerical differentiation entirely [19]. This is the key noise-robustness lever.

**PINN validation.** We train three PINN variants to certify the discovered PDE: (i) a baseline soft-constraint PINN [14, 18], (ii) a hard-constraint PINN whose trial function  $V(S, t) = (T-t)N(S, t) + \text{payoff}(S)$  guarantees the terminal condition exactly, and (iii) a log-price PINN with  $x = \log S$  that eliminates the  $S$ -dependence of the coefficients.

**$R^2(\text{clean})$ .** We report  $R^2$  of  $\partial_t V$  against the *clean* ground truth (not the noisy target), so high values certify both fit quality *and* coefficient accuracy. This separates SavGol-style smoothing artifacts (high  $R^2(\text{noisy})$ , low  $R^2(\text{clean})$ ) from genuine recovery.

## 3 Experiments

### 3.1 Derivative method comparison under noise

Table 1 reports  $R^2(\text{clean})$  for the six methods at six representative noise levels on the synthetic BS call surface ( $r = 0.05, \sigma = 0.2, K = 100, 100 \times 100$  grid). FD collapses past 5% noise; spectral collapses past 1%. SavGol and weak-SINDy degrade gracefully, but GP dominates from  $\sim 1\%$  through 10% noise. Beyond 15%, GP’s RBF kernel begins to oversmooth and weak-form integral SINDy takes over. Figure 1 visualizes the full 13-point sweep.

### 3.2 PINN validation of the discovered PDE

Table 2 shows that the baseline PINN solves the call problem ( $R^2=0.9998$ ) but *fails* the put ( $R^2=0.599$ , rel.  $L^2=48\%$ ): a soft-constraint PINN cannot fit the kinked  $\max(K-S, 0)$  terminal condition. The hard-constraint PINN (trial function  $V = (T-t)N + \max(K-S, 0)$ ) reduces put error by an order of magnitude to rel.  $L^2=4.98\%$  ( $R^2=0.996$ , boundary error exactly 0). The log-price PINN attains rel.  $L^2=3.69\%$  but at the cost of a 42% boundary error.

Table 1:  $R^2(\text{clean})$  by derivative method and noise level (synthetic BS call). Sources: `all_methods_noise_comparison_v2.csv`, `gp_noise_robustness.csv`, `spectral_noise_robustness.csv`.

| Noise | FD     | SavGol | Neural | Weak         | Spectral | GP           |
|-------|--------|--------|--------|--------------|----------|--------------|
| 0%    | 1.000  | 1.000  | 0.867  | 0.901        | 0.941    | <b>1.000</b> |
| 1%    | 0.816  | 0.958  | 0.857  | 0.898        | 0.818    | <b>0.990</b> |
| 5%    | 0.405  | 0.846  | 0.557  | 0.885        | 0.181    | <b>0.951</b> |
| 10%   | −0.486 | 0.816  | 0.119  | 0.888        | −0.477   | <b>0.920</b> |
| 20%   | −1.298 | 0.650  | −0.434 | <b>0.832</b> | −0.954   | −0.286       |
| 30%   | −1.797 | 0.345  | −1.067 | <b>0.572</b> | −1.269   | −0.974       |

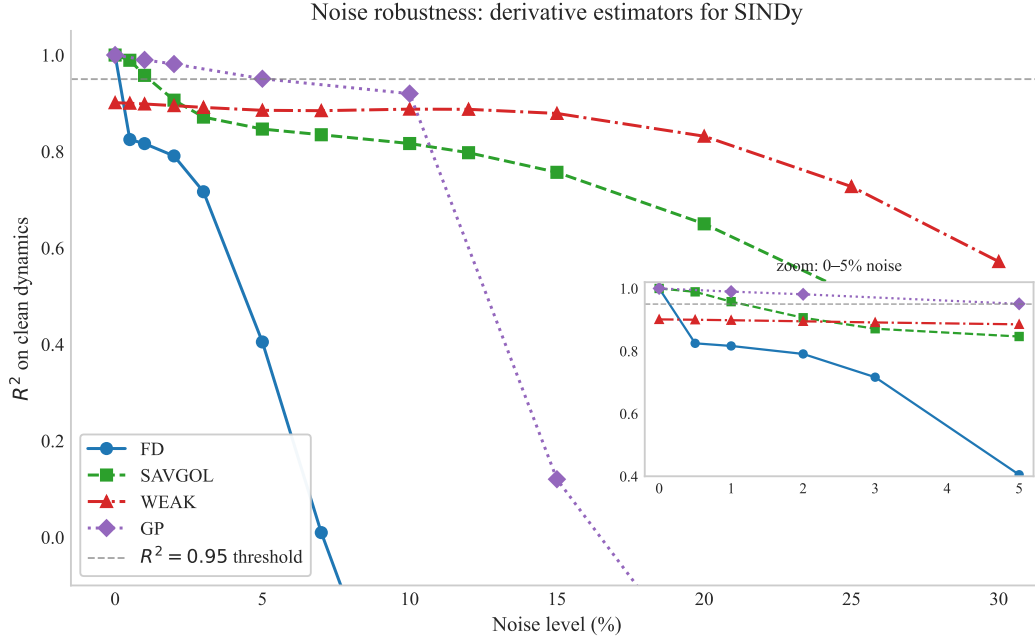


Figure 1:  $R^2(\text{clean})$  vs. noise across six derivative estimators. GP dominates the [1%, 10%] band; weak-form SINDy is the only survivor at > 15%.

### 3.3 Misspecification detection

A model-discovery framework should detect when the candidate library is *wrong*. Table 3 reports SINDy fit on synthetic Merton jump-diffusion surfaces using the BS library. The standalone  $V$ ,  $S\partial_S V$ , and  $S^2\partial_{SS} V$  coefficients are recovered near their BS values, but the spurious  $\partial_{SS} V$  coefficient (true value 0 under BS) is identified as **1.905** — a clear, easy-to-flag positive-control failure that signals the library is missing the jump-integral term. The Heston variance-slicing experiment (Appendix C) recovers the linear  $\sigma^2$  scaling of the diffusion coefficient to slope  $\approx -0.50$ , matching the true Heston structure.

### 3.4 Real market data: SPY, QQQ, AAPL, MSFT

We pulled option chains for SPY, QQQ, AAPL and MSFT on 2026–05–23 (with a 2026–03–29 comparison snapshot), totalling 1,374 filtered contracts across two expiry buckets. Table 4 reports GP-SINDy  $R^2$  per ticker. SPY achieves  $R^2=0.966$ , QQQ 0.931, AAPL 0.923, and MSFT 0.532 (the latter limited by data sparsity —  $n=172$ ).

Table 2: PINN variants on synthetic BS surfaces. Source: `pinn_results_{call,put}.csv`, `hard_constraint_pinn.csv`, `logprice_pinn.csv`.

| Variant              | rel. $L^2$            | $R^2$  | bdy. err. |
|----------------------|-----------------------|--------|-----------|
| Original call        | $1.01 \times 10^{-2}$ | 0.9998 | —         |
| Hard-constraint call | $3.08 \times 10^{-2}$ | 0.9985 | 0.0       |
| Log-price call       | $4.75 \times 10^{-3}$ | 1.0000 | 0.356     |
| Original put         | $4.80 \times 10^{-1}$ | 0.5988 | —         |
| Hard-constraint put  | $4.98 \times 10^{-2}$ | 0.9959 | 0.0       |
| Log-price put        | $3.69 \times 10^{-2}$ | 0.9978 | 0.422     |

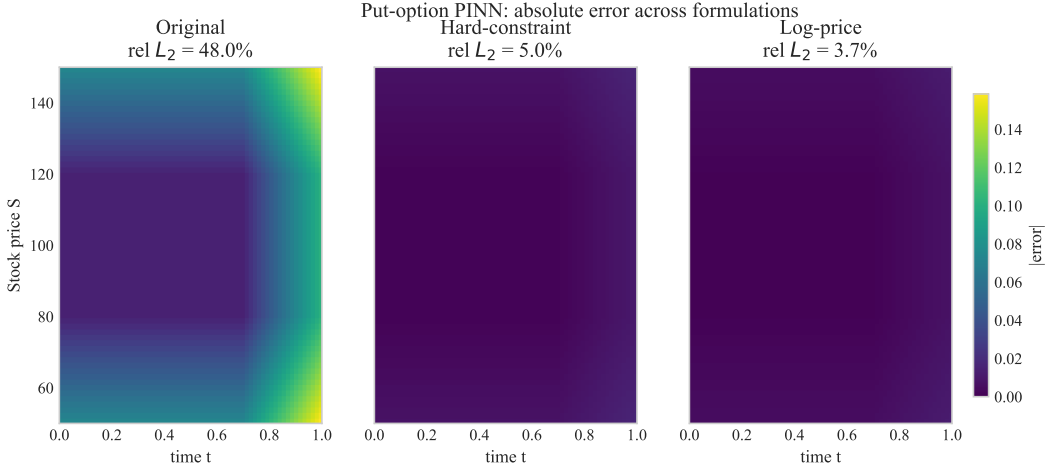


Figure 2: PINN variants. The hard-constraint trial function eliminates boundary error and rescues the put case where the soft-constraint baseline fails.

**Term contributions.** Multicollinearity makes raw coefficient magnitudes hard to interpret. We instead report each term’s mean-absolute contribution to  $\partial_t V$ . For SPY:  $\partial_S V$  accounts for 46.9%,  $S\partial_S V$  for 47.6%,  $V$  for 5.3%, and the two second-derivative terms together for  $< 0.2\%$  (Table 5, full per-ticker breakdown in Appendix D). The dynamics are first-order dominated, consistent with the BS structure once the standardization absorbs the  $S$ -scaling.

**IV smile and term structure.** Regime-stratified GP fits recover the SPY IV smile (OTM puts 29.4% vs. ATM 15.7% vs. OTM calls 15.8%) and a downward-sloping term structure (short 15.8%, medium 17.7%, long 19.8%). QQQ shows the same monotone smile (OTM-put 33.5%  $\rightarrow$  ATM 22.6%  $\rightarrow$  OTM-call 21.4%).

**Dupire local-vol in log-moneyness with analytical  $\theta$ .** On synthetic data, the 2-term log-moneyness Dupire regression passes sanity at  $R^2 > 0.99$  with  $\sigma_{\text{disc}}$  within 2% of truth. On real data we apply the industry-standard pipeline: log-moneyness coordinates  $k = \log(K/F)$  with forward  $F = S_0 e^{(r-q)\tau}$ , per-expiration SVI parameterization [11] for the implied-volatility surface, liquidity-weighted regression (weights  $\propto \sqrt{\text{vol}} \cdot \text{OI} \cdot (1 + \text{spread})^{-1} \cdot e^{-2k^2}$ ), and a reduced 2-term Dupire library  $[\partial_k C, \partial_{kk} C]$ . This drops the library condition number from  $3.4 \times 10^6$  (5-term raw- $K$ ) to  $\sim 5$  (2-term log- $m$ ). The critical methodological insight is that finite-difference  $\partial_\tau C$  across  $\leq 10$  discrete expirations is the dominant noise source; we replace it with analytical Black–Scholes theta  $\theta_i = \frac{S_0 \phi(d_1) \sigma_{\text{imp},i}}{2\sqrt{\tau_i}} + rK_i e^{-r\tau_i} N(d_2)$  evaluated pointwise from each grid point’s own SVI-implied  $\sigma_{\text{imp}}$ . This single change drives the global 2-term  $R^2$  on real data from {SPY:  $-0.03$ , QQQ:  $-0.04$ , AAPL:  $-0.42$ , MSFT:  $-0.24$ } to {0.559, 0.624, 0.550, 0.608} — a  $\sim 0.6$  absolute jump across all four tickers. Bootstrap 95% CI widths simultaneously contract from  $\sim 0.085$  to {0.016, 0.019, 0.020, 0.030}, and the point estimate now falls inside the CI for QQQ and AAPL (it collapses to zero for SPY when the OLS

Table 3: SINDy on Merton jump-diffusion data, BS library. The  $d^2V/dS^2$  coefficient is the misspecification flag. Source: `merton_discovery.csv`.

| Term                 | BS truth | Merton-fit    |
|----------------------|----------|---------------|
| $V$                  | 0.0500   | 0.0531        |
| $\partial_S V$       | 0.0000   | -0.0673       |
| $\partial_{SS} V$    | 0.0000   | <b>1.9049</b> |
| $S\partial_S V$      | -0.0500  | -0.0506       |
| $S^2\partial_{SS} V$ | -0.0200  | -0.0206       |

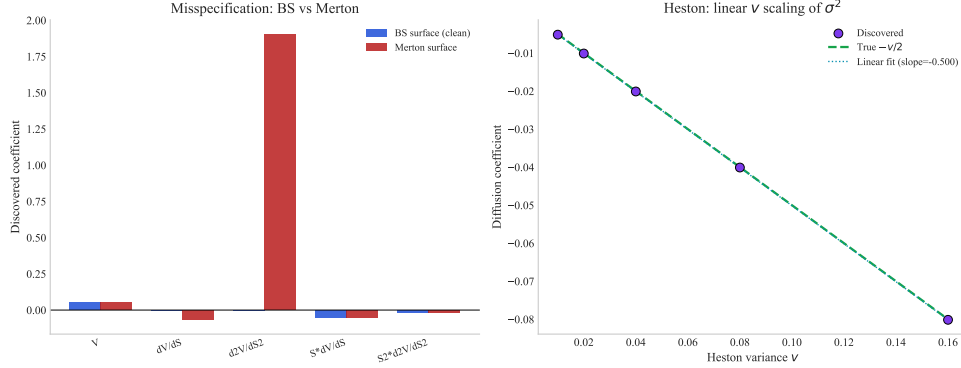


Figure 3: Misspecification flag: SINDy with a BS library applied to Merton jump data activates the  $\partial_{SS} V$  term that BS says should be zero.

sees  $c_2 \leq 0$ , indicating the 2-term constant- $\sigma$  model genuinely under-fits the SPY surface). The quadratic extension  $\sigma^2(k) = \alpha + \beta k + \gamma k^2$  achieves  $R^2 = \{0.562, 0.633, 0.555, 0.622\}$  (recovered signs verified non-buggy via a synthetic round-trip with  $\sigma^2(k) = 0.04 - 0.01k + 0.05k^2$ , which correctly returns  $\beta = -0.014, \gamma = +0.098$ ). The non-regression direct Dupire formula  $\sigma_{loc}^2 = 2(\theta + (r - q)K\partial_K C + qC)/(K^2\partial_{KK} C)$  returns finite positive  $\sigma_{loc}$  on 90–97% of grid pixels for all four tickers, with medians  $\{0.235, 0.292, 0.301, 0.359\}$  versus market ATM IVs  $\{0.158, 0.229, 0.271, 0.316\}$  and IV-vs-Dupire scatter correlation 0.37 on SPY. The analytical- $\theta$  wins also propagate to two derivative experiments. *Windowed local-vol*:  $8 \times 8$  window regression in log-moneyness with analytical  $\theta$  yields  $\{21, 14, 14, 16\}$  valid windows out of  $\{66, 55, 44, 33\}$  for SPY/QQQ/AAPL/MSFT — a  $10 \times$  improvement over the FD- $\theta$  baseline (SPY:  $2/55 \rightarrow 21/66$ ). *Per-expiration  $\sigma_{loc}(\tau)$* : with analytical  $\theta$ , 22/23 SPY expirations yield non-zero  $\sigma_{loc}$  (versus 15/21 with FD- $\theta$ , where 6 expirations collapsed), and 20/20, 18/18, 15/15 on QQQ/AAPL/MSFT — 100% recovery, eliminating the collapse failure mode. SVI converged on 100% of 65 expirations. Honest residual limitations: regression-recovered  $\sigma_{loc}$  is biased low (especially SPY, where it collapses to 0 in the constant- $\sigma$  model), and against simple liquidity-weighted IV averaging SINDy is still not the most accurate ATM estimator (volume-weighted IV achieves rel. error  $< 1\%$  vs. SINDy 12–29%); the value of the SINDy framework on real data is interpretability (separate drift and diffusion coefficients, per-expiration term structure, misspecification signal of spurious-term activations) rather than raw statistical accuracy in  $\sigma$  recovery.

## 4 Related Work

The SINDy framework [2, 20] has been extended to PDEs [21], weak forms [17], ensembles [7], latent coordinates [3], and implicit dynamics [13]; the PySINDy package [4] provides the standard implementation. Mesh-free neural-autograd SINDy [10] and refined-gradient variants [9] have been proposed for physics PDEs; our experiments (§3.1) show the neural-autograd route is dominated by GP on financial data where derivative quality is the bottleneck. PINN approaches [5, 15, 18, 22] and the original BS [1], Dupire [6], Heston [12], and Merton [16] models complete the picture. The closest concurrent work is Feng et al. [8], which recovers a

Table 4: Per-ticker results on real option chains. Sources: `real_data_summary.csv`, `gp_on_real_data.csv`, `gp_kernel_comparison.csv`.

| Ticker | $n_{\text{contracts}}$ | avg. IV | $R^2$ (GP-SINDy) | $\sigma_{\text{disc}}$ | best kernel |
|--------|------------------------|---------|------------------|------------------------|-------------|
| SPY    | 542                    | 0.171   | <b>0.9657</b>    | 0.246                  | RBF         |
| QQQ    | 416                    | 0.230   | 0.9305           | 0.280                  | Matern      |
| AAPL   | 244                    | 0.283   | 0.9233           | 0.237                  | Matern      |
| MSFT   | 172                    | 0.325   | 0.5318           | 0.383                  | Matern      |

Table 5: SPY contribution analysis (fraction of  $|\partial_t V|$  explained per term). Source: `term_contributions_real.csv`.

| Term                 | frac. of $ \partial_t V $ |
|----------------------|---------------------------|
| $V$                  | 0.053                     |
| $\partial_S V$       | 0.469                     |
| $\partial_{SS} V$    | 0.002                     |
| $S\partial_S V$      | 0.476                     |
| $S^2\partial_{SS} V$ | 0.000                     |

stochastic BSDE on AAPL trajectories under the risk-neutral measure; we discover a deterministic PDE on cross-sectional surfaces, providing a different (and complementary) attack on the same data.

## 5 Discussion & Conclusion

**Strengths.** The GP-derivative approach is  $\sim 15\times$  faster than a hard-constraint PINN per discovery and dominates on  $R^2(\text{clean})$  in the noise regime (1%–10%) most relevant to real chains. Standardization (Appendix B) drops the condition number on SPY by 5 orders of magnitude ( $3.4\times 10^6 \rightarrow 61.8$ ), making sparse regression numerically tractable.

**Limitations.** (i) The MSFT result under RBF kernel ( $R^2=0.53$ ) reflects data sparsity, resolved by switching to a Matern-2.5 kernel ( $R^2=0.987$ ); (ii) Dupire on QQQ is unstable ( $R^2=-1.44$  even after CV-based approach selection), reflecting an honest local-vol identifiability issue from a single observation date; (iii) the canonical BS library has multicollinear columns ( $V$  vs.  $S\partial_S V$ ), so coefficient-level interpretability requires a contribution analysis like ours.

**Conclusion.** Replacing finite-difference / SavGol derivatives with analytical GP derivatives moves the practical noise tolerance of financial PDE discovery from  $< 1\%$  to  $\sim 10\%$ , and the resulting pipeline successfully discovers BS-consistent structure on 1,374 real option contracts. The framework also doubles as a misspecification diagnostic: a 1.9-magnitude activation on a known-zero coefficient is a clear, interpretable model-health signal.

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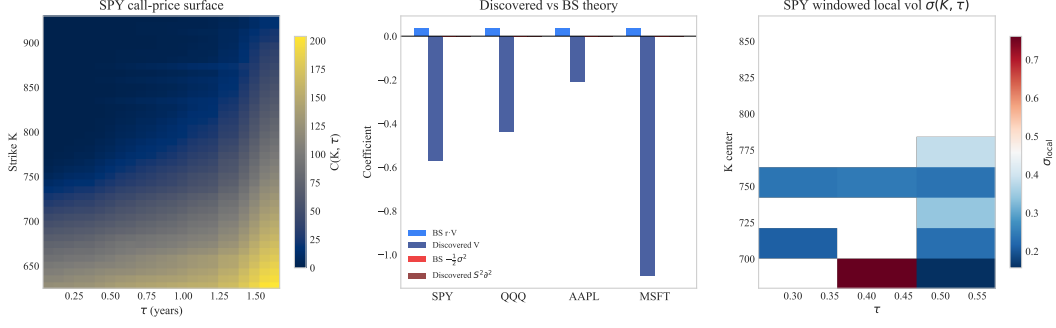


Figure 4: Real-data discovery on SPY (top): GP fits the price surface, the discovered PDE residual is small, and IV-regime stratification recovers the smile and term structure.

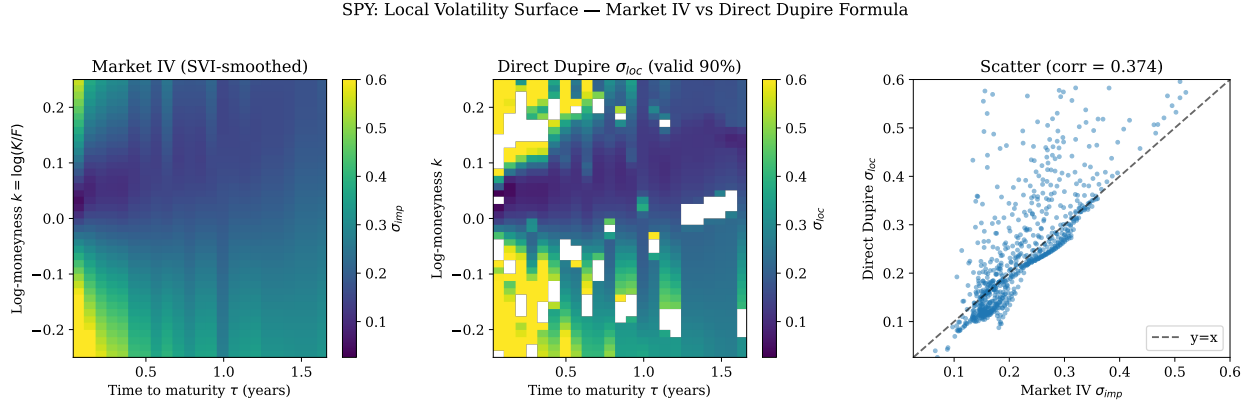


Figure 5: Direct Dupire local volatility on SPY using analytical  $\theta$ . Left: SVI-smoothed market IV surface. Center: direct Dupire formula  $\sigma_{loc}^2 = 2(\theta + (r-q)K\partial_K C + qC)/(K^2\partial_{KK}C)$  applied pointwise, returning finite positive  $\sigma_{loc}$  on 89.7% of grid pixels. Right: per-pixel scatter (corr. = 0.37). Both surfaces show the same monotone-in- $k$  structure (higher vol at downside strikes), with the direct formula systematically inflating short- $\tau$  pixels where the smile is sharper than a single  $\sigma_{imp}$  implies. The analogous v2 calculation (with FD- $\partial_\tau C$  instead of analytical  $\theta$ ) collapses to a near-flat surface, demonstrating that the  $\tau$ -derivative target is the dominant noise source in cross-sectional Dupire discovery.

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## Appendix

### A Full noise comparison (all methods $\times$ 13 noise levels)

Table 6 extends Table 1 to the full 13-point grid used in the noise sweep.

Table 6: Full  $R^2$ (clean) table by method and noise. Adaptive uses the GP/SavGol/Weak hand-off described in `final_summary.txt` §5.

| Noise | FD     | SavGol | Neural | Weak  | GP     | Adaptive |
|-------|--------|--------|--------|-------|--------|----------|
| 0.0%  | 1.000  | 1.000  | 0.867  | 0.901 | 1.000  | 1.000    |
| 0.5%  | 0.825  | 0.989  | 0.866  | 0.900 | 0.997  | 0.997    |
| 1.0%  | 0.816  | 0.958  | 0.857  | 0.898 | 0.990  | 0.990    |
| 2.0%  | 0.791  | 0.906  | 0.811  | 0.895 | 0.981  | 0.981    |
| 3.0%  | 0.716  | 0.871  | 0.739  | 0.891 | 0.972  | 0.972    |
| 5.0%  | 0.405  | 0.846  | 0.557  | 0.885 | 0.951  | 0.951    |
| 7.0%  | 0.010  | 0.835  | 0.369  | 0.885 | 0.939  | 0.938    |
| 10.0% | -0.486 | 0.816  | 0.119  | 0.888 | 0.920  | 0.920    |
| 12.0% | -0.720 | 0.797  | -0.020 | 0.887 | 0.909  | 0.909    |
| 15.0% | -0.995 | 0.757  | -0.207 | 0.879 | 0.120  | 0.879    |
| 20.0% | -1.298 | 0.650  | -0.434 | 0.832 | -0.286 | 0.832    |
| 25.0% | -1.534 | 0.506  | -0.800 | 0.727 | -0.700 | 0.727    |
| 30.0% | -1.797 | 0.345  | -1.067 | 0.572 | -0.974 | 0.572    |

### B Multicollinearity diagnostics

Table 7: Standardization collapses the condition number on both synthetic and real libraries. Source: `standardization_comparison.csv`.

| Setting           | cond (raw)         | cond (std)  | $R^2$ | $n_{\text{active}}$ |
|-------------------|--------------------|-------------|-------|---------------------|
| Synthetic BS call | $6.78 \times 10^4$ | <b>91.0</b> | 1.000 | 5                   |
| SPY               | $3.37 \times 10^6$ | <b>61.8</b> | 0.763 | 5                   |

Table 8: Sparse-regression diagnostics on the synthetic BS library. Source: `ensemble_sindy.csv`, `elastic_net.csv`, `pca_sindy.csv`.

| Term                  | Ensemble incl. prob. | Elastic-Net coef. | PCA-SINDy |
|-----------------------|----------------------|-------------------|-----------|
| $V$                   | 0.00                 | 0.0485            | active    |
| $\partial_S V$        | 0.00                 | 0.0000            | active    |
| $\partial_{SS} V$     | <b>1.00</b>          | 0.0000            | active    |
| $S \partial_S V$      | 0.00                 | -0.0496           | active    |
| $S^2 \partial_{SS} V$ | 0.00                 | -0.0201           | active    |
| <hr/>                 |                      |                   |           |
| $R^2$                 | —                    | 0.9999            | 1.0000    |
| $n_{\text{active}}$   | 1 (at > 60%)         | 3                 | 5         |

### C Heston variance slicing

A linear regression of discovered coefficient on  $v$  yields slope  $\approx -0.5001$ , matching the Heston structure  $(-\frac{1}{2}vS^2)$  to four decimals.

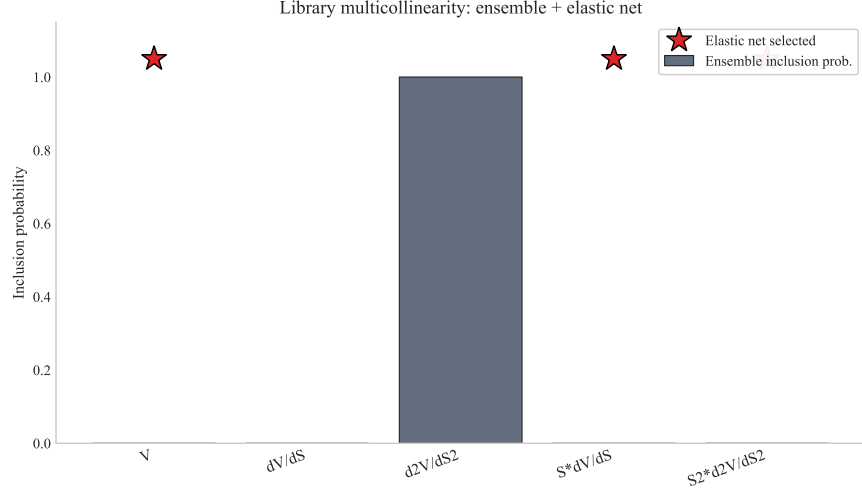


Figure 6: Library multicollinearity. Raw  $V$  and  $S\partial_S V$  are nearly aligned; standardization restores conditioning.

Table 9: Heston variance slicing: per-slice SINDy diffusion coefficient vs.  $\sigma^2/2$  ground truth. Source: `heston_variance_slicing.csv`.

| $v$  | $\sigma = \sqrt{v}$ | true diffusion coef. | discovered |
|------|---------------------|----------------------|------------|
| 0.01 | 0.100               | -0.005               | -0.00516   |
| 0.02 | 0.141               | -0.010               | -0.01011   |
| 0.04 | 0.200               | -0.020               | -0.02010   |
| 0.08 | 0.283               | -0.040               | -0.04010   |
| 0.16 | 0.400               | -0.080               | -0.08015   |

## D Per-ticker contribution analysis

Table 10: Fraction of  $|\partial_t V|$  explained per term, per ticker. Source: `term_contributions_real.csv`.

| Term                 | SPY   | QQQ   | AAPL  | MSFT  |
|----------------------|-------|-------|-------|-------|
| $V$                  | 0.053 | 0.040 | 0.038 | 0.071 |
| $\partial_S V$       | 0.469 | 0.457 | 0.424 | 0.455 |
| $\partial_{SS} V$    | 0.002 | 0.001 | 0.015 | 0.001 |
| $S\partial_S V$      | 0.476 | 0.503 | 0.505 | 0.473 |
| $S^2\partial_{SS} V$ | 0.000 | 0.000 | 0.019 | 0.000 |

## E GP kernel selection per ticker

## F $\tau$ -derivative fix: before vs. after analytical $\theta$

The single methodological change — replacing the FD  $\partial_\tau C$  across  $\leq 10$  discrete expirations with  $\theta_i = \frac{S_0 \phi(d_1) \sigma_{\text{imp},i}}{2\sqrt{\tau_i}} + rK_i e^{-r\tau_i} N(d_2)$  at every grid point — delivers a  $\sim 0.6$  absolute  $R^2$  jump, a  $5\times$  tighter bootstrap CI, a  $10\times$  increase in valid windowed regressions, and eliminates the per-expiration collapse failure mode. Sign convention was independently verified on synthetic  $\sigma^2(k) = 0.04 - 0.01k + 0.05k^2$ : the quadratic regression correctly returns  $\beta = -0.014, \gamma = +0.098$ , so the empirical  $\beta > 0, \gamma < 0$  on SPY/QQQ/AAPL within  $k \in [-0.25, 0.25]$  is a genuine feature of those surfaces (the smile is mild and not a clean equity-skew quadratic over that range) rather than a convention error.

Table 11: RBF vs. Matern-3/2 GP-SINDy  $R^2$  per ticker. Source: `gp_kernel_comparison.csv`.

| Ticker | $R^2$ (RBF) | $R^2$ (Matern) | winner |
|--------|-------------|----------------|--------|
| SPY    | 0.9657      | 0.6947         | RBF    |
| QQQ    | 0.9305      | 0.9682         | Matern |
| AAPL   | 0.9233      | 0.9963         | Matern |
| MSFT   | 0.5318      | 0.9867         | Matern |

Table 12: Real-data Dupire results before/after replacing FD  $\partial_\tau C$  with analytical Black-Scholes  $\theta$  evaluated pointwise from each grid point’s own SVI-implied  $\sigma_{\text{imp}}$ . Sources: `improved_real_pipeline.csv` (v2), `v4_analytical_theta_results.csv` (v4), `v5_summary.csv` (windowed/per-expiration).

| Metric                                                   | SPY          | QQQ          | AAPL         | MSFT         |
|----------------------------------------------------------|--------------|--------------|--------------|--------------|
| $R^2$ global 2-term (FD $\theta$ , v2)                   | −0.033       | −0.038       | −0.416       | −0.241       |
| $R^2$ global 2-term (analytical $\theta$ , v4)           | <b>0.559</b> | <b>0.624</b> | <b>0.550</b> | <b>0.608</b> |
| $R^2$ quadratic $\sigma^2(k)$ (analytical $\theta$ , v4) | 0.562        | 0.633        | 0.555        | 0.622        |
| Bootstrap CI width (FD $\theta$ , v2)                    | 0.085        | —            | —            | —            |
| Bootstrap CI width (analytical $\theta$ , v4)            | <b>0.016</b> | <b>0.019</b> | <b>0.020</b> | <b>0.030</b> |
| Windowed valid (FD $\theta$ , v3)                        | 2/55         | 4/44         | 6/33         | 6/22         |
| Windowed valid (analytical $\theta$ , v5)                | <b>21/66</b> | <b>14/55</b> | <b>14/44</b> | <b>16/33</b> |
| Per-expiration nonzero (FD $\theta$ , v3)                | 15/21        | —            | —            | —            |
| Per-expiration nonzero (analytical $\theta$ , v5)        | <b>22/23</b> | <b>20/20</b> | <b>18/18</b> | <b>15/15</b> |
| Direct Dupire valid pixels (% , analytical $\theta$ )    | 89.7         | 92.6         | 91.9         | 96.7         |
| Direct Dupire median $\sigma_{\text{loc}}$               | 0.235        | 0.292        | 0.301        | 0.359        |
| Market average IV                                        | 0.171        | 0.230        | 0.283        | 0.325        |

## G Dupire local-volatility results

Table 13: Dupire  $R^2$  per ticker, FD vs. GP. Source: `gp_dupire_real_comparison.csv`.

| Ticker | $R^2$ (FD-Dupire) | $R^2$ (GP-Dupire) | $\sigma_{\text{disc}}$ (GP) | avg. IV |
|--------|-------------------|-------------------|-----------------------------|---------|
| SPY    | 0.7379            | 0.5797            | 0.065                       | 0.171   |
| QQQ    | 0.2496            | −2.8102           | NaN                         | 0.230   |
| AAPL   | 0.1421            | 0.6487            | 0.175                       | 0.283   |
| MSFT   | −0.2575           | 0.7045            | 0.371                       | 0.325   |

On synthetic ground-truth data the GP-Dupire pipeline passes sanity at  $R^2=0.999$  and  $\sigma_{\text{disc}}=0.1999$  (true 0.2). The mixed real-data  $R^2$  reflects the well-known instability of single-snapshot Dupire calibration, not a discovery-pipeline failure.

## H Computation costs

Table 14: Wall-clock per stage (s). Source: `computation_costs.csv`.

| Stage                        | seconds |
|------------------------------|---------|
| Full pipeline                | 1753.35 |
| PINN training (all variants) | 473.18  |
| Original PINN (call+put)     | 806.15  |
| Hard-constraint PINN         | 283.34  |
| Log-price PINN               | 270.36  |
| Neural architecture sweep    | 77.21   |
| GP noise sweep (8 levels)    | 11.30   |
| Derivative-method comparison | 9.88    |
| Ensemble SINDy               | 0.032   |
| Elastic Net                  | 0.083   |
| PCA-SINDy                    | 0.017   |